Weathering and Erosion

Introduction

The processes of weathering and erosion work together to transform the Earth through their natural processes. The processes operate over extended periods to modify all types of natural features. The processes create soil and various landforms which scientists observe in the present day.

Weathering

The natural environment of Earth experiences weathering through which rocks undergo a process that divides them into smaller fragments.

The process of weathering occurs at a specific location which generates materials that will be transported through erosion.

Types of Weathering

1 Phyiscal (Mechanical) Weathering

This type breaks rocks into smaller pieces without changing their chemical composition.

The process occurs through:

- Temperature changes (expansion and contraction)
- Ice wedging
- Pressure release

2 Chemical Weathering

This process occurs when rocks undergo chemical alterations because of their contact with water and oxygen and acids.

Common examples include:

- Oxidation (rusting)
- Acid rain reacting with limestone

3 Biological Weathering

This type of weathering occurs because living organisms create disruption through their activities.

The following activities create this weathering:

- Through tree roots which invade cracks
- Through burrowing which animals perform
- Through microorganisms

Erosion

The process of erosion enables the movement of weathered materials from one site to another.

Erosion differs from weathering because it involves the active movement of material.

Agents of Erosion:

- Water – Rivers, rainfall, ocean waves
- Wind – Moves sand and fine particles
- Ice – Glaciers scrape and carry rocks
- Gravity – Causes landslides and rockfalls

The process of weathering causes rock to disintegrate while the process of erosion transports the broken material.

Deposition

Erosion moves material to a different location where the material will settle down. The operation which achieves this function operates through the process of deposition.

The following examples demonstrate the process:

- River deltas
- Sand dunes
- Beaches

Conclusion

The ongoing processes of weathering and erosion and deposition work together to transform the Earth's surface. The combination of these natural processes creates soil while they shape the Earth's surface.

Thank You !